Shenyao Jin

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Education

Colorado School of Mines, Reservoir Characterization Project — Golden, Colorado
Ph.D. Geophysics — *Aug 2023 – May 2027*

- Advisor: Dr. Ge Jin, Committee Chair: Dr. Hossein Kazemi
- **Focus**: *Seismic inverse problems, distributed fiber-optic sensing (DFOS), inversion algorithms, numerical modeling, and optimization for subsurface characterization*
- **Minor:** *Computer Science*

Zhejiang University — Hangzhou, China
B.S. Geoscience — *Sept 2019 – Aug 2023*

- **Thesis**: "Clustering-Based Joint Inversion for Imaging Shallow Subsurface Geological Targets"
- Thesis Advisor: Dr. Zhanjie Shi

Technical Skills

- **Programming:** Python, MATLAB, C/C++, LaTeX, Bash; proficient in Linux/Unix environments
- **Software and Packages**: Seismic Un*x, MOOSE (Multiphysics Object-Oriented Simulation Environment), DASCore (Distributed Acoustic Sensing data processing), Devito (seismic wave propagation modeling), Scikit-learn, TensorFlow
- **Inverse Problems & Optimization**: Gauss-Newton iteration, gradient-based optimization, regularization techniques, Tikhonov inversion, history matching, parameter estimation
- **Geophysical Methods:** Seismic imaging and inversion, DFOS/DAS data processing and modeling, strain-pressure analysis, hydraulic fracture interpretation, forward modeling, signal denoising, numerical wave propagation
- **Machine Learning & AI**: Supervised learning for geophysical inversion, neural network surrogate models, machine learning-enhanced data interpretation
- **Related Coursework:** Inversion Theory, Advanced Machine Learning, Reservoir Simulation, Advanced Geophysical Computing, Advanced Data Science, Numerical Methods
- **Software Development:** Author of fibeRIS, an open-source Python package for history matching and interpretation of DAS data (github.com/shenyaojin/fibeRIS).

Experience

Geophysical Imaging Intern — Houston, Texas
TGS — *May 2026 – Aug 2026*

- Designed a machine learning-based denoising workflow for 4D DAS VSP data from ExxonMobil's Guyana dataset, suppressing tube waves and flow noise that limit time-lapse (4D) interpretation.
- Trained two specialist DUCK-Net (PyTorch) noise-prediction models and built a banded pipeline routing each channel band to its best model via f-k diagnostics; reduced tube-wave energy from 88% to 1.4% while preserving reflection signal (12.8× selectivity).
- Scaled the workflow to a 144,257-shot field volume through MDIO data conversion and containerized SLURM jobs on HPC clusters; delivered registered models and a fully reproducible f-k QC framework.

Graduate Research Assistant — Golden, Colorado
Reservoir Characterization Project, Colorado School of Mines — *Aug 2023 – Present*

- **Cement Quality Monitoring using Low-Frequency DAS (ExxonMobil Project Mariner)**
  - Developed a low-frequency DAS (LF-DAS) data processing workflow and field deployment strategy to monitor cement quality in horizontal wells.
  - Implemented a coupled pressure-strain numerical model using MOOSE to simulate LF-DAS forward responses under various cementing conditions.
  - Conducted iterative history matching of field LF-DAS data to validate methodology effectiveness in identifying cement channeling.
- **Permeability Inversion from DAS Data (ExxonMobil Collaboration)**
  - Formulated and solved an inverse problem for low-frequency DAS data to quantify in-situ permeability; developed physics-based forward modeling and optimization algorithms for inversion.
  - Advanced inverse problem solving through iterative history matching of field datasets, demonstrating methodology effectiveness for subsurface characterization.
  - Integrated analytical and physics-based approaches with optimization techniques to validate inversion results against well logs and operational data.

- **Geothermal Energy Systems Data Interpretation (Fervo Energy Collaboration)**
  - Collaborating with Rice University and Fervo Energy to develop inversion-driven interpretation tools for multiple geophysical data types (e.g., DAS, sonic, seismic).
  - Designing novel machine learning-enhanced inversion workflows for LF-DAS data to analyze pressure-strain coupling and fracture characterization.

Graduate Research Intern Idaho Springs, Colorado

Reservoir Characterization Project, Colorado School of Mines *May 2025 – Aug 2025*

- Conducted DOE-funded research on monitoring induced fracture propagation using low-frequency DAS; developed novel data acquisition and processing strategies.
- Designed and deployed a novel U-shaped fiber-optic cable in monitoring wells with co-located DTS and DSS sensing fibers, advancing field deployment methodologies.
- Analyzed multi-channel DAS datasets to extract fracture signatures and validate subsurface imaging workflows.

Undergraduate Research Intern Golden, Colorado

Reservoir Characterization Project, Colorado School of Mines *July 2022 – September 2022*

- Developed data processing and inversion workflows for low-frequency DAS data collected in Lake Hattie, Wyoming; implemented velocity scanning and optimization algorithms for enhanced data interpretation.
- Contributed to research workflow development for DAS-based subsurface imaging and applied signal processing techniques to improve data quality.

Undergraduate Research Assistant Hangzhou, China

School of Geosciences, Zhejiang University *Sept 2021 – June 2023*

- Developed advanced joint inversion algorithms in *C++* to solve coupled seismic and electromagnetic inverse problems for subsurface characterization.
- Implemented optimization techniques including Gauss-Newton iteration and regularization methods to improve inversion stability and convergence.
- Validated inversion algorithms on field datasets and well log data, demonstrating practical effectiveness in resolving subsurface geological structures.

Abstracts and Presentations

Shenyao J., Ge J. *Low-Frequency DAS for Cement Quality Monitoring in Horizontal Wells*

- URTeC 2026, Houston, Texas. Oral Presentation.
- SEG-AAPG IMAGE 2025, Houston, Texas. Oral Presentation.
- American Geophysical Union Fall Meeting 2025, New Orleans, Louisiana. Oral Presentation.
- RCP Sponsor Meeting 2025, Golden, Colorado. Oral Presentation.

Shenyao J., Ge J. *Conductive Fracture Monitoring Using Distributed Strain Sensing: From Stimulation to Production*

- SEG-AAPG IMAGE 2024, Houston, Texas. Poster Presentation.
- RCP Sponsor Meeting 2024, Golden, Colorado. Oral Presentation.

Publications

Jin, S., & Jin, G. (2026). Low-frequency DAS for cement quality evaluation in horizontal wells. In *Proceedings of the SPE/AAPG/SEG Unconventional Resources Technology Conference*, Houston, Texas, USA. URTEC-4459984-MS. https://doi.org/10.15530/urtec-2026-4459984

Jin, S., & Jin, G. (2026). Low-frequency DAS for cement quality monitoring in horizontal wells. *SPE Journal* (in review).

Mao, F., Yang, B., & **Jin, S.** (2025). Recovering missing regions of earth magnetic anomaly grid data (EMAG2) using RePaint based on diffusion model. *Big Data and Earth System*, 1(1), 100004.

Honors and Awards

George R. Pickett Memorial Scholarship Oct 2024

- Awarded for excellence in borehole geophysics with applied focus on oil and gas workflows.

First-Year Fellowship for Graduate Students Mar 2024

- Awarded by the Colorado School of Mines to recognize incoming students for integrating computational geophysics to advance sustainable energy systems.

Meritorious Winner, Mathematical Contest in Modeling (MCM) Feb 2022

- Top 8% of international undergraduate teams. Contributed as lead programmer and data analyst.